



Aerodynamic Optimization of Pyramid Stacking for Container Ships Through CFD

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Abstract

Over the recent decade, container ships have gone through a dramatic boom in size, while attention spans have been paid toward hydrodynamic characteristics rather than aerodynamic features. This lack popped up when a 20000-TEU container ship of Ever Green line went aground under a gale in the Suez channel for six days and somewhat the sea transportation, and subsequently, the global economy was captivated. Despite few studies on mega-sized container ship stacking configurations, the aerodynamic features of pyramid stacking types have not ever been studied in particular. To this end, the present paper aims to unravel the case by testing various cross-sectional pyramid types' layouts in a set of 2D simulations using the realizable k- ϵ turbulence model through CFD commercial software, STARCCM+. Further, the outcomes have been processed for combining pyramid profiles in 3D models to study frontal and lateral drag. The outcome of the study is twofold; the first is to show the extension of 2D simulation applicability for evaluating the aerodynamic behavior of different stacking profiles for 3D scenarios. Second, simulations demonstrated that various pyramid stacking could effectively reduce the lateral drag up to 22% compared to the full-load configuration while preserving loading capacity. As such, the optimum pyramid layout is finally presented, whereas it could be applied in loading instruction to enhance mega-container ship interoperability and maneuvering efficiency.

Keywords: Container ship, aerodynamic, Computational Fluid Dynamic (CFD), pyramid stacking, drag

1. Introduction

Over the last two decades, the container ship size has gone through a soar in size from 6000 TEU in 1996 to 24000 TEU in 2021 [1]. In the meantime, the world fleet also has tripled, and the share of container trade in the

global economy also has gone through a sharp surge to over 2000 million tons in 2019 [2]. This trend has highlighted the role of the mega-container ship in the world economy far more than before, so studies on mega-sized container ships have been kicked off since ten years ago. Compared to the other types of vessels, container ships carry over 50% of cargo on deck [3], so it makes them unique from an aerodynamic features standpoint. Further to the negative impact on fuel efficiency, wind load affects the maneuverability of container ships in the constrained waterway and shallow waters. Recently, on 23rd March 2021, the evergreen, a mega-size container ship with 20000 TEU loading capacity went aground in Suez Canal and has blocked one of the most important arteries of global trade for 6 days (please see figure 1), while the sequential impact of late cargo delivery to the costumers jeopardized the world trade by pushing up the prices of commodities even up to the end-user [28].



Figure 1. Blockage of evergreen mega container ship in Suez Canal [4] (left), and a container ship with pyramid stacking configuration (right).

Although many hydrodynamic factors have a role to play in the operability of a ship in the restricted waterways such as bank effect and squat, the role of lateral force on the mega-size containership is indispensable. Based on local authorities, the vessel was under the west-to-east gale that pushed the vessel at the time of the incident [5]. In a nutshell, the wind force could be one of the main key players that affected the maneuverability of evergreen in this incident, and through a wide lens, in



safety, maneuverability, operability, fuel efficiency, wear and tear of steering gear system of the all container ships.

Historically, the researches about wind load force on huge container ships back to experimental tests in the wind tunnel [6] and [7], where the impact of different container stacking on wind load was studied. As per outcomes, the random-loaded ship experienced higher longitudinal force and less lateral force. Following that, extensive container ship stacking configurations were scrutinized in wind tunnel tests for aerodynamic features by [8]. A 9000+ TEU container ship model was studied in a set of configuration scenarios in a 360° angle of attack (AOA) in terms of frontal and lateral drag. In fact, the pyramid stacking configuration was investigated for the first time in Andersen's study to identify aerodynamic features in three different models. The term "Pyramid Stacking" reckons a practice of omitting outermost containers in upper tiers to mitigate the risk of losing containers due to roll motions. In this loading practice, the planar area of a stack is reduced in every succeeding upper tier as depicted in figure. 1, whereas this method has been used in Maersk and other shipping lines to lessen extensive loads to container lashing equipment and lower the risk of rupture in case of roll motion. According to [8], the pyramid stacking could decrease the lateral loads in 90° AOA up to 12.5 % compared to the full-stack configuration. Following Andersen, Majidian et al [9] modeled only one case of pyramid stacking in 0° AOA through CFD, and the results agreed with Anderson's data. Up to now, these two studies have been the only researches regarding pyramid stacking on a container ship. Although the studies about the general configuration of container stacking contribute the lion's share of the literature in this agenda, the details of the configuration and its pertinent impacts on lateral drag, in particular, have been neglected. As a result, the present research aims to give an insight into the influence of different

pyramid stacking layouts on the lateral drag of huge container ships and introduce an optimal pattern in this issue.

After a concise introduction, section 2 is supposed to examine 10 different pyramid profiles through 2D simulation and propose the optimum profiles for 3D studies. Section 3 goes into the details of 3D simulation and presenting the obtained outcomes. Section 4, as the fruit of the present study, is going to discuss CFD outputs and introduce the best practice for loading purposes. Ultimately, all outcomes are summarized in the section 5, as the juice of study.

2. 2D simulation of the stacking profiles

Before starting the 3D simulation, simulation of 2D profiles can give an insight into the story to converge the scenarios into specific cases that can save computational cost and effort. On the other hand, the positive outcomes could be a model for future extension of the study of the same ilk. However, this fact should not be neglected that apart from flow behaviors in the third ordinate, 2D simulation is short of indicating fluid-particle interaction spatially. Therefore, the 2D simulation result must be taken with a grain of salt, and they could be only a preface for further investigation.

2.1. Model

Given the incident and high windage area of EverGreen, this ship has been chosen for study as a sample. The principal particulars of the model are given in Table. 1, and for 2D simulation, the profiles followed the exact real-world dimension. Using the same dimension for the 3D model could facilitate future investigation, even though the scaled model was used owing to some intrinsic limitations in the software. The CAD model is processed to the commercial CFD package of STARCCM+. The reason behind using this software is inspired high capability of the solver for fast

and accurate processing [10] and [11], and also due to the user-friendly interface compared to the other commercial packages.

Table 1. Principal Dimension of M/V Ever Green [12].

Item	Figure
Tonnage (GRT)	220940
Tonnage (DWT)	199629
Displacement	265876 t
LOA (Length overall)	399.94 m
Beam	58.8 m
Draught (max)	16.0 m
Depth	23.9 m
Speed	22.8 Kn

The proposed 2D profiles for simulation are depicted in the figure 2, while the number of reduced containers varies from 6 to 12 for the whole bay. The size and dimension of 40 feet containers are adapted as per [13]. The symmetric shape of the profile has been used to facilitate the interpretation of flow development around the profile. Besides, the cross-sectional plane is picked from the center of the ship in LOA/2 that is assumed to be the main spot of wind effect.

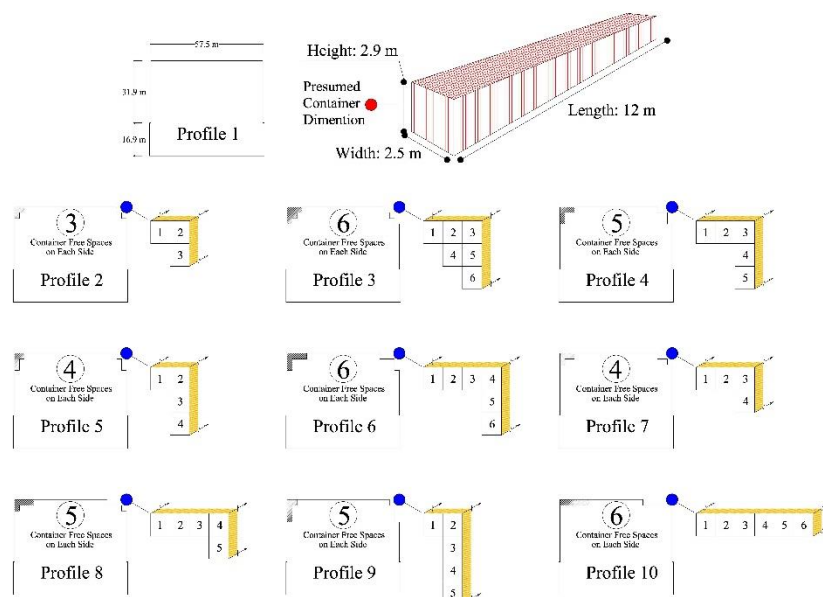


Figure 2. An indication of 2D profile for container bays.

Several assumptions have been considered for CFD simulation:

- The ship is supposed to be upright, even keel, and the ballast condition is not considered in this study, so given the objective, only a fixated freeboard has been adapted.
- Admittedly, in real-word, the acting wind load vector on the vessel is the resultant vector of reverse ship motion vector and wind vector, even though this fact is not considered in this study and the simulation was carried out in the static condition.
- Despite the transient nature of wind flow, it has been considered steady and incompressible.
- As it will be explained, holding the Reynolds similarity between vessel and model is hard to elaborate, so the simulation has been compromised partially to safeguard the feasibility of results.

2.2. Governing equations

Navier Stokes equation, as the mother formula for prediction of flow dynamism, is the governor equation in this setting. Since the direct solution of the equation is relatively impossible, the recursive approaches are being used to present a fair approximation of the results for calculating averaged momentum equation in the iterative scheme. The solver in StarCCM+ utilized both average continuity and momentum equations to estimate the velocity and pressure fields recursively for the fluid, so the readers are referred to the main source for further details [14] and [15].

$$\frac{\partial(\rho \bar{u}_i)}{\partial x_i} = 0 \quad (1)$$

$$\frac{\partial(\rho \bar{u}_i)}{\partial t} + \frac{\partial}{\partial x_i} (\rho \bar{u}_i \bar{u}_j + \rho \bar{u}_i' \bar{u}_j') = - \frac{\partial \bar{P}}{\partial x_i} + \frac{\partial \bar{\tau}_{ij}}{\partial x_i} \quad (2)$$

$$\bar{\tau}_{ij} = \mu \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \quad (3)$$

In the above equation, U_i is the time-averaged velocity, $\bar{P}$ is the time-averaged pressure, μ is viscosity, and ρ is the density. In these settings, i, j assigns the direction of flow, $\bar{\tau}_{ij}$ is the tensor of mean viscosity, and finally, $\overline{\rho u_i u_j}$ denotes Reynold stress tensor. The answer for this term could be derived by modeling the term of turbulent viscosity μ_t , known as eddy viscosity. Various RANS-based turbulence models have come to propose a scheme for modeling turbulent viscosity, where the K-Epsilon model suits flow with a high Reynolds number and few separations of the flow. Based on the Boussinesq hypothesis, equation. 4 estimates the term of Reynold stress, even though the term of eddy viscosity μ_t should be modeled first through approximating two transport partial differential equations (PDEs). In the following equation, S is the mean rate of the strain tensor, K is turbulence kinetic energy, Δ is the Kronecker delta, C_μ is constant, and ε is the rate of dissipation. The downside of the model is in isotropic assumption in terms of eddy viscosity, though it is functional for the sake of this research.

$$-\overline{\rho u_i u_j} = 2\mu_t S_{ij}^* - \frac{2}{3}\rho k \delta_{ij} \quad (4)$$

$$\mu_t = C_\mu \rho \frac{K^2}{\varepsilon} \quad (5)$$

Realizable k-epsilon has been adapted for either 2D and 3D simulation. The perk of the Realizable K-Epsilon model is the inclusion of a new transport equation for dissipation rate for estimating Reynolds stress tensor [16]. In addition, the critical coefficients are not constant, and they are set as the function of mean flow and turbulence properties. Besides, considering the high Reynolds number in the simulation and separation regions over the model, the realizable model suits for generating turbulence phenomenon [17]. In addition, a two-layer approach for the K-Epsilon model that is utilized in StarCCM+ solver is used, and audiences are referred to [15] for further information. The Finite Volume Method

(FVM) is utilized in the solver to split up the computational domain into small grids, and through applying the initial values and boundary conditions, the mass conservation and momentum equations are solved to derive a linear system of differential equations for modeling velocity and pressure fields.

2.3. Definition of drag coefficient

Since only 90 degrees AOA is the subject of the present study, the ship coordinate system matches the global coordinate system as a standard Cartesian right-handed coordinate system. The ship coordinate system is located right in midship in the half of LOA, Z-axis is positive upward, X-axis is positive forward, and the Y-axis is positive to the port side. Since the ship coordinate system is fixated based on a global coordinate, so wind flows in - X-direction. Figure 3 depicts the coordinate system for the ship model in the domain.

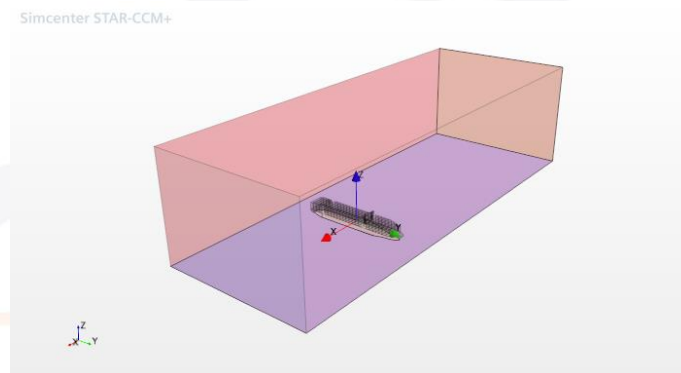


Figure 3. An indication of coordination system.

Considering the objective of the study, only the force in the X direction is subjected to measurement. For making the result independent from ship size and wind velocity, it should be non-dimensional, so equation 4 denotes the non-dimensional coefficients. Here, X is the force in N, ρ is the density in kg/Cu.m, U is the mean flow velocity, and A_f is the frontal area.



$$C_x = \frac{x}{\frac{1}{2} \cdot \rho \cdot U^2 \cdot A_f} \quad (6)$$

2.4. CFD Set up, and outcomes

The details of CFD simulation for 2D profiles are given in table 2. Two incentives were the primary drivers for 2D simulation; the first is the successful former experiences [18] and [19] that provided an estimation of physical phenomenon for further applications. The second is saving computational cost and time plus having an insight into the cases before getting deeper into 3D simulations. Admittedly, the 2D simulation may by no means indicate some reliable outcomes for many 3D simulations. Considering the complex nature of turbulence, the flow patterns in interaction with other bays are of a wider zoom, and the 2D simulation is not inclusive in this issue.

The trim mesh that has been utilized for 2D and 3D modeling substantially improves the mesh metric and the flow of data. Only the sea force 6, with a velocity of 12.6 m/s has been chosen as one of the most dominant sea forces [20]. Before starting the simulation, a square with the ship size has been modeled to validate CFD outputs. Apart from grids density and extension, the application and implementation of wall function have a significant role in capturing the flow near the wall and boundary layers. Frequently, Y^+ is offered to be between 30 and 70 [21], while for the sake of the present simulation, 50 has been chosen. Besides, the ratio of 1/2 for the transition from one layer to another has been kept in the whole domain. According to [22], the drag coefficient for a 2D square has been reported to be 2.05, whereas the CFD setup demonstrated the result by only a 2.1% difference on 2.0926, so the applied condition can be confidently used.

Table 2. Mesh and boundary condition applied in 2D domain.

Mesh		Boundary Condition	
Item	specification	Inlet	Velocity inlet
Total Number	Between 2,100,000 to 2,300,000	Outlet	Pressure Outlet
Y plus	50	Roof	Velocity inlet
Reynolds number	25,102,702	Bottom	Symmetry

The same setup was adapted for 2D profiles. The sequence of grid size and extension is presented in the figure 4. The obtained drag coefficients are normalized to facilitate the comparison of the models. As shown, the $Y+ < 50$ has been perfectly held over the profile, and all models indicated the same results. (see figure 4 for profile 3.)

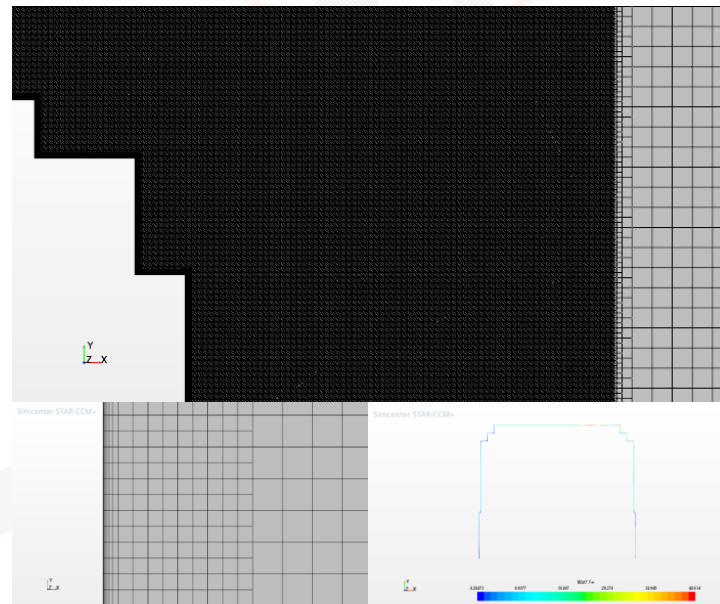


Figure 4. The mesh grid density and extension of grid on boundary layer for 2D profiles, and The presentation of $Y+$ on the surface of the profile (bottom-right).

Based on the figure 5, the 2D simulation entailed surprising outcomes. Despite authors' expectation, profile 1 as the full-stack condition stood among the low-coefficient group and made it potentially nominated as a case study for further 3D simulation, where the former experimental and CFD are available for comparison purposes. Unlike [8], profile 1, indicated

less drag coefficient compared to profile 2 and 3. Nevertheless, drawing a comparison between profiles 2 and 3, the results agreed to the experimental trend. As a result, and since the experimental data is available as per [8], profiles 2 and 3 are nominated for further investigations. On the other hand, profile 6 demonstrated the minimum value out of data, while it is followed in the next stage by profile 7. Since the number of loaded containers in each bay in profile 7 is more than in profile 6, it carries a higher loading capacity and is worth studying. Hence, out of the proposed models, five profiles have been chosen for further 3D investigations.

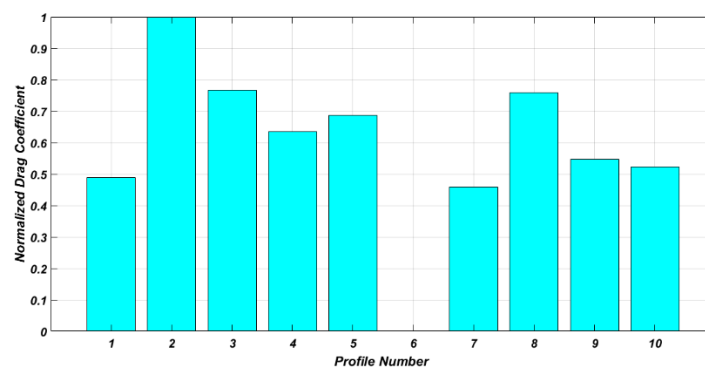


Figure 5. The result of 2D simulations for profiles in figure 3.

Regardless of the force and further 3D results, the 2D simulations have a significant point to offer. According to the figures reducing one container transversely is much more beneficial than vertically, where a comparison between profile 5, and profile 7, could positively confirm this hypothesis. Additionally, the formation of low-pressure regions between the containers reduces the chance of separation on the upper surface of the tiers, so that mitigates the turbulence and consequently drag force as it is depicted in the figure 6.

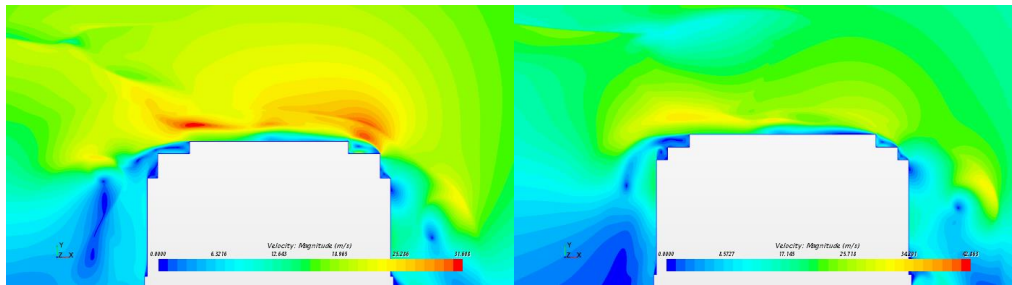


Figure 6. The indication of velocity contours over profile 6 (left) and 7 (right). The wind flows in – X-direction.

As previously mentioned, the outputs of 2D simulations will be processed to 3D models in order to reveal other angles of the case. Up until the 2D results have not shown very reliable unless the reason behind discrepancy could be found. The 3D simulation in the next section not only can be used for real-world cases, but it shows to what extent 2D simulation for the present application could be informative.

3. 3D simulations

This section goes to delineating the nuances of the CFD setup. At first, the 3D setup is given in detail to provide reproducibility of the model, then the correction factor and uncertainty of results are concisely described. Finally, the validated results will enable us to divulge the CFD outcomes.

3.1. Simulation set up

The 3D simulation has dwelled on previous literature to adjust the setup. The water and its pertinent interaction are excluded from the simulation. The domain size is set up according to [17], as is illustrated in the figure 7. Since the models abound with sharp edges, the number of meshes has been dramatically increased to cover up the edges and to fill the gap between bays. Plus, an additional refinement region around the body bolstered the modeling of the fields in near far areas, off the bays, and capturing the wakefield behind the bay. The details of boundary conditions and grid setup are summarized in Table 3, and audiences are

referred to [9], [3], and [17] for more details. Given the complexity of the geometries, the number of cells varied between 16 to 17 x 10⁶. Since the results have not significantly changed using six inflation layers, to save computational costs, the number of prism layers is set to be 6. It ought to be noted, this condition could be only held in case the number of meshes outside the prism layer does not experience a jump or high transition to guarantee the linearity between cells output. In this case, since the trimmed cells have been perfectly applied in a cubic shape with a very high aspect ratio of almost 1, this setup can be valid (see figure 7).

Table 3. Mesh and boundary condition applied in 3D domain.

Boundary Conditions				
Inlet	Velocity 12.6 m/s			
Outlet	Pressure outlet			
Sea surface	No-slip wall			
Ship	Wall			
Y plus	50			
Absolute working pressure	1 bar			
Air density @ 25 c	1.168 Kg/ Cu.m			
Reynolds Number	4,870,424			
Based on the Mach number (0.036), no changes in density (incompressibility), and not having time derivative in momentum equation, segregated pressure-velocity solver has been chosen				
Mesh Specifications				
Prism layer thickness	0.015 m			
First Layer height	0.0021m			
Surface mesh size	0.06 m			
Maximum size on the far domain	6.0 m			
Refinement Region	0.12 m			
Mesh Metrics				
minFaceValidity	minCellQuality	minFacePlanarity	minVolumeChange	minSkewness
1.000000	0.000099	0.809271	0.012668	0.022251

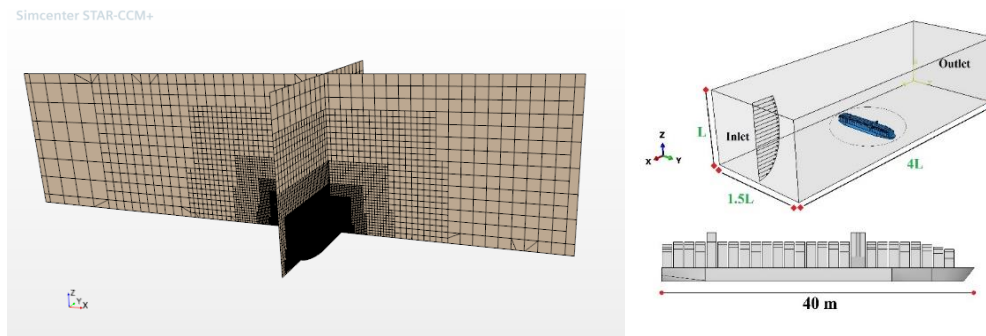


Figure 7. The layout of the mesh over the surface and between container bays (left), the extension of the grid for the whole computational domain (right), and the computational domain data (bottom).

3.2. Blockage factor

According to the guidelines for simulation of the virtual wind tunnel, blockage ratio was taken into account that has been suggested to be less than 5% for experimental test and less than 3% for CFD analysis [23], while for directional blockage factor, this value should be held less than 17% [24]. The formulation of blockage ratio is simple, where they are just the ratio of the model frontal and lateral area to the wind tunnel directional or cross-section. The blockage ratio is not the subject of the present study, since the lateral force matters. The directional blockage ratio here is 66%, so to mitigate the blockage impact of the model, the blockage correction factor according to [25] has been adapted like Andersen's experiment [8].

3.3. Grid independency and sensitivity analysis

Based on the recommended verification method by [26], the numerical uncertainty USN comprises three factors; iterative convergence uncertainty UI, grid-spacing uncertainty UG, and time-step uncertainty UT as per equation. 7. On this basis, and since the simulation is steady-state, the iterative convergence uncertainty UI is negligible, and grid-spacing uncertainty UG is of attention.

$$U_{SN}^2 = U_I^2 + U_G^2 + U_T^2 \quad (7)$$

To that end, the grid-spacing uncertainty has been carried out according to the Richardson extrapolation principle through three global mesh setups with a refinement ratio of $r_G = \sqrt{2}$. For more details, the readers are referred to [26] and [27]. The results of uncertainty analysis are given in Table. 4. The data has been computed for the case of full-stack since the non-dimensional experimental data is available. The associated uncertainty has to be considered in this setting, even though the study demonstrated the robustness of mesh set up by only less-than-2-percent uncertainty.

Table 4. Mesh convergence and uncertainty study for drag coefficient.

Full Stack Loading	S1	S2	S3	RG	δ_{REG1}^*	UG(%S1)
Drag Coefficient	1.290931	1.296619	1.299747	1.82	0.012	1.9

3.4. Simulation results

The CFD results have been corrected by the blockage factor, but the uncertainty has not been applied. Furthermore, the CFD simulation is performed by 1:10 scaled model since the software showed an error for using the ship's actual dimensions. In this order, two factors are chief; Reynold similarity and Y^+ . In terms of Reynolds similarity, the velocity has to be scaled up in the reverse order for the scaled model. However, in this simulation, as previously addressed, the wind speed was set at 12.6 m/s to avoid probable errors, and that is why the outcomes of the CFD do not necessarily reflect the results for the actual ship. In fact, CFD in this setting aims to give an insight for further experimental study and examine the idea that to what extent different stacking profiles play role in lateral drag. As a result, the outcomes can only predict and reflect the difference of drag coefficient for different stacking configurations. Since the experimental results are non-dimensional, still the CFD outcomes can be employed for

comparison purposes. According to figure. 8, profile 7 indicates the minimum drag coefficient out of simulated profiles. Surprisingly, profile 7 is also optimum in comparison with profile 3, as a usual configuration in loading practice while keeping more loading capacity. Totally, and compared to profile 1, it seems pyramid configurations in any form have the upper hand in aerodynamic efficiency. Last but not least, 2D outcomes demonstrated to be far from 3D results in details, however, in a broader look, the trend match to some extent, and the probable causes behind that will be contemplated in the discussion section.

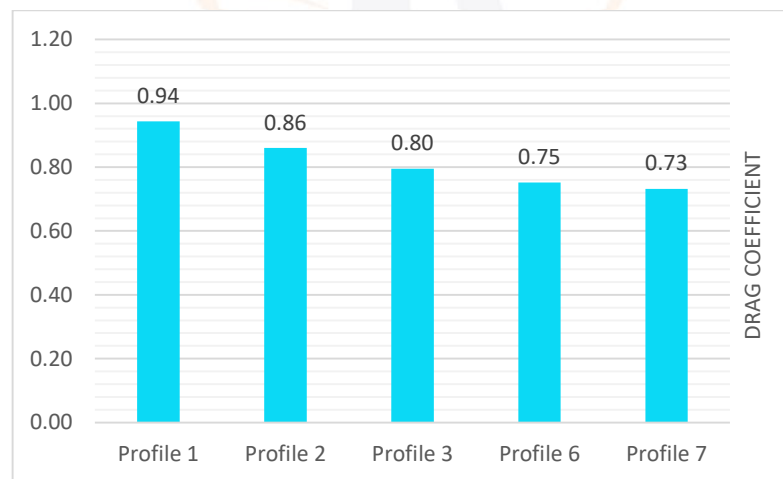


Figure 8. Drag coefficient for different profiles.

As previously mentioned, the point of using CFD in this study is to take some steps further beyond existing knowledge, even though this task essentially demands some piece of experimental data to benchmark CFD outcomes. So, the results have been benchmarked with [8], and the comparison is presented in Table 5. As can be seen, there is a striking resemblance between CFD and experimental data, particularly in full-stack conditions by less than 1% difference. The CFD data also well-agreed profiles 2 and 3 by the maximum difference of 6.8% for profile 3. Although the overall agreement is very fair, divergence can be observed toward profile 3. That is to predict, the difference between other stacking profiles

with experimental data might be higher with respect to the complexity of geometry, though within an acceptable range. Given the computational expenses, the K-ε model predicts the phenomenon quantitatively quite well. From the validation of data, it can be concluded that the CFD setup has been functional, and the results for other loading scenarios are expected to be estimated informatively.

Table 5. Difference between CFD and Experimental Data [8].

$\text{Difference} = \frac{\text{CD Exp} - \text{CD CFD}}{\text{CD Exp}} \times 100$ Percentage difference	Profile 1	Profile 2	Profile 3
	0.94%	2.6%	6.8%

4. Discussion

To sum up, three objectives are of the focus in this section; the first is to understand that to what extent 2D simulations are informative about real-world data. The second is to find out the impact of different pyramid stacking configurations on lateral drag by keeping eyes on the velocity and pressure field over the containers. Ultimately, three new random stacking configurations, resemble the real-world loading, will be studied. As the fruit of objectives, some prescriptions and guidelines about the optimum models will be brought up in the end.

4.1. 2D Vs. 3D Models

Considering the applications of 2D models in the aerodynamics context, they were intended to alleviate the endeavor of simulations. However, as expected, the 2D simulation could not reflect the story perfectly, although some points are worth mentioning. Using 2D models not only cut down the extension of eddies in third ordinates but also does not essentially consider the transfer of momentum in the eddies between container bays. For comparison, the velocity field in cross-section of profile 7 in 3D and 2D simulations is shown in the figure 9.

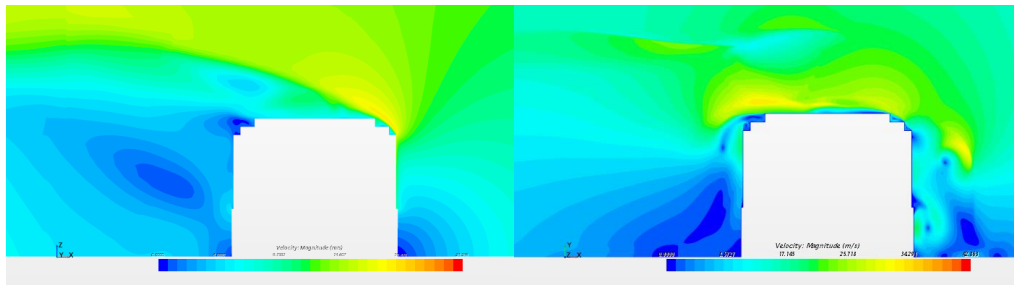


Figure 9. Velocity field in cross section of profile.7, left picture 3D and right picture for 2D. Note the wind flow runs from right to left.

As can be witnessed, the difference between velocity fields is observable downstream and upstream. Because the gap between containers lets the flow streams, while in the 2D scheme, the flow diffracts from the surface and consequently perturbs the wind profile before encountering the surface - almost, four regions of swirling can be detected with different sizes in this area. This limitation affected the downstream through bounding the flow as well, and thus the low-velocity region did not show up behind the bay. The large high-pressure area behind the bay in the 3D simulation leads to the transfer of momentum between the flow layer once it leaves the posterior containers. Technically, the 2D simulation only predicted the separation on the outermost containers, where the containers are reduced, and that is the most significant similarity in velocity field between 2D and 3D simulation. By and large, 2D modeling can give an insight into the boundary layer phenomena since the foundation of Y^+ and wall function is set based on a 2D flow profile. Even though this is the area where 2D and 3D simulations have become close, this does not perfectly reflect the real-world flow behavior since Y^+ , in its essence, is compromised for simplicity. Nevertheless, the boundary layer interaction is the overlap between 2D and 3D simulations.

4.2. Flow Characteristics

The velocity field over the containers comprises notable information regarding the behavior of the flow in different stacking configurations, even though at the onset, admittedly, the cross-sectional pictures only reflect the velocity field in midship. By drawing a comparison between the velocity field in profiles 1 and 7 as the maximum and minimum drag, many differences can be spotted that explain the reason behind drag coefficients (see figure 10). Apparently, the flow in profile 1 is much more chaotic, and eddies instigated by each bay have intertwined together. As a result, the flow has lost its energy and speed, so the sequence of interaction between vortex packets formed a very wide region of the low-velocity field behind the model. The drag force directly depends on the size of these vortices; the larger eddies are, the higher is drag force. Hence, the interaction between bays carries a crucial role in increasing turbulence and giving the surge to drag force. The difference in turbulence intensification can be clearly observed in the space between bays, in the upper part of all bays, where it is shown by the red box in the figure 10. Therefore, the shape of outermost tiers positively streamlined the flow and impeded the formation of eddies in profile 7.

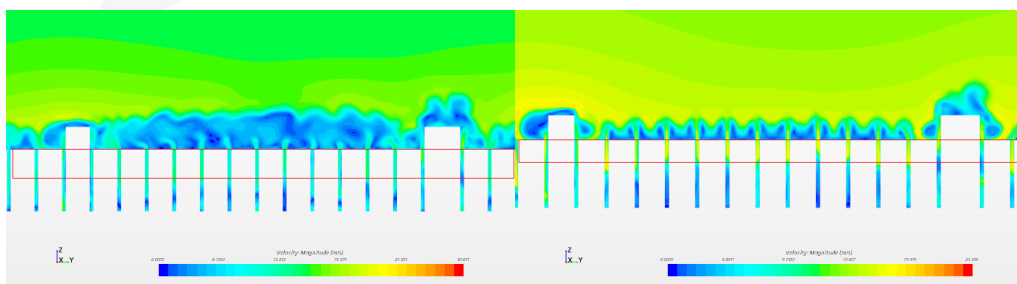


Figure 10. Velocity field over the containers profile. 1 left, and profile. 7 right.

As can be seen in the figure 11, the tips of the arrows are spotting the regions where vortices between bays have become intermixed, and that explains why profile 3 showed a higher drag value. Additionally, the

pressure field in the utmost aft and forward bays brings about two intense low-pressure areas in the lateral wind. This fact can spark this idea that apart from turbulence over the bays in midship, optimization of the bays on the fore and aft deck may have a role to play in reducing lateral drag.

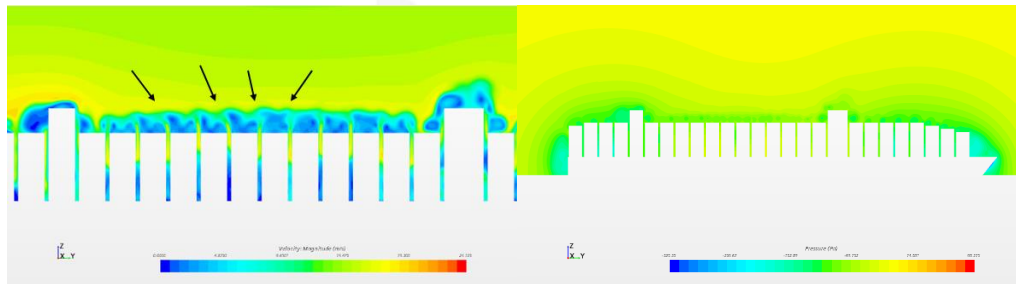


Figure 11. The cross section of velocity fields over the containers profile. 3 left, and the pressure field in profile. 7.

4.3. Mixed Configurations

The final objective of the research is to bring the analytical information into the practical instructions, so this sub-section is going to bridge between findings and application to introduce the optimum solution. Refer to Table 6, profile 7 decreased the lateral drag up to 22.5 % while preserving loading capacity by reducing 192 containers compared to the full-load capacity. Although profile 6 indicates drag reduction of 20.4%, profile 7 is a trade-off between loading capacity and drag reduction. In a nutshell, all pyramid models regardless of the shape of profile have positively influenced drag coefficient.

Table 6. The information about drag coefficient and reduced containers in each 3D configuration.

Profile	Reduction in drag coefficient relative to profile. 1 %	Number of reduced containers
2	8.9%	144
3	15.7 %	288
6	20.4 %	288
7	22.5 %	192
R1	15.8%	246
R2	13.4%	258
R3	20.1%	96

Nevertheless, the story in real-world scenarios is quite different. These models can be employed in the first loading port since a mega-sized container ship may unload many containers in the ports of call in a row, so the stacking configurations would be arbitrary then. Therefore, three random stacking configurations have been modeled in addition to mimic the real-world cases. However, the symmetric bays are considered not only for simplification but to open the results for interpretation. To this end, and based on [8] and [9], tries have been directed to introduce two models that are relatively streamlined in fore and aft deck to improve frontal drag as well. The models are as follows, according to the figure 12:

- R1: The configurations are a randomized combination of profiles 1 to 10.
- R2: Foredeck is streamlined by profile. 3, aft deck by profile. 6, and the middle deck is randomized by a combination of profiles 1 to 10.
- R3: To have a tradeoff between loading capacity and reducing lateral drag in low-pressure regions that have been addressed, the middle bays are loaded by profile 1, and fore and aft by profile 7.

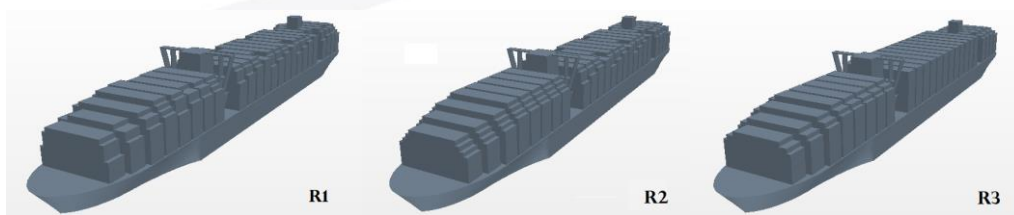


Figure 12. The randomized 3D configurations.

Based on the obtained results, R3 showed considerable results with 20.1% drag reduction, while the minimum number of the container has been omitted compared to all configurations. That is to say, the speculation about the flow behavior in the fore and aft deck pressure field has been valid. So, in a nutshell, two loading configurations could be introduced to



be the most optimum models; loading container ship using profile 7, and R2 model. Nevertheless, in terms of a broader prescription for loading in departure, apart from using these models, some pieces of advice can be suggested;

- The streamlining of the fore and aft deck should be the priority since it can accelerate the wind flow in these areas that are far from midship and potentially prone to increase the force moment and influencing maneuverability.
- Even in the pyramid shape, reducing containers horizontally is of favor, or at least, the equal number of containers should be reduced in both directions.

5. Conclusion

Given the significance of wind load on the huge container ships, a mega-sized container ship was selected for studying different loading configurations with pyramid profiles. To save computational cost and having a preview before implementing 3D simulations, 10 different pyramid profiles have been subjected to 2D simulation. In the next step, the 3D simulations of the vessel have been carried out based on the 2D outcomes. Although 2D simulation opened a window to the task, the results have not been comprehensibly informative regarding the flow behavior on cross-sectional profile, but boundary layer to some extent. However, 3D simulations have offered invaluable upshots for reducing lateral drag. Accordingly, the lateral drag coefficient effectively decreased up to 22.5% compared to the full-load stacking configuration in profile 7. Looking closer to the flow behavior over the containers, and given the real-world loading considerations, three mixed loading configurations have been further investigated, while the results showed an optimum configuration cut down the lateral drag up to 20.1% with only 96

containers omission. Consequently, the following instruction can be drawn for operational decision-makers.

1. Pyramid stacking in any symmetric layout reduces lateral wind load.
2. Reducing containers horizontally from the utmost tier is much fruitful than vertically.
3. Regardless of the number of the deck for loading, the attempts should be focused to streamline bays on the fore and aft deck.
4. Profile. 7 is much more efficient than the profile 2 and 3 that are regular loading practices at the moment.
5. Ultimately, the R3 model can be the best trade-off between loading capacity and optimum lateral wind drag.

6. Future works

The following milestones can be portrayed for future works:

- Increasing the range of AOAs for simulation to obtain the efficiency of the optimum profiles in this paper for other AOAs.
- Dynamic simulation of wind load using the overset method in StarCCM to model the vessel drift due to wind load.
- Finally, simulation of single bay stacking and comparison of results with 2D simulation to deduce the influence of other bays on the flow. These results could be fruitful for further structural studies on the tension and loads on the lashing equipment.

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